ESILV - Tutorial #5

Commodity Markets and Models

- This session (worth **5%**) lasts only 1h30.
- Solve exercises **on paper** and hand it in to your tutorial leader (*one per group*) before the end. (no email) – (no late submission)
- Ask your instructor for advice. Corrections will not be posted on DVL.
- In case of absence to this tutorial, you will be graded as zero. **Your instructor does not need to know the reason of your absence.** Official absence needs to be recorded directly with the scholar office (`scolarite-esilv@devinci.fr`).

Problem 1: `Regime Jumps in Electricity Prices`

This tutorial exercise is based on the regime-switching model for electricity prices introduced by the reference article:

- Huisman, R., & Mahieu, R. (2003). Regime jumps in electricity prices. *Energy economics*, 25(5), 425-434.

You will analyze a complete implementation that simulates electricity spot prices from 2010 to 2026, estimates model parameters via maximum likelihood, and produces diagnostic outputs. The exercise tests your understanding of:

- The mathematical formulation of regime-switching models
- Economic interpretation of model parameters
- Analysis of simulation outputs and parameter estimates
- Comparison with alternative modeling approaches

Mathematical Framework

Log-Price Decomposition

Following Huisman and Mahieu (2003), the natural logarithm of the electricity spot price $s(t)$ is decomposed as:

$$s(t) = f(t) + x(t), \quad t = 1, \ldots, T \tag{1}$$

where $f(t)$ is the deterministic component and $x(t)$ is the stochastic component.

Deterministic Component with Weekend Effects

$$f(t) = \mu_0 + \beta_1 D_1(t) + \beta_2 D_2(t) \tag{2}$$

where:

- μ_0: Equilibrium log-price level
- $D_1(t) = 1$ if day t is Saturday, 0 otherwise

- $D_2(t) = 1$ if day t is Sunday, 0 otherwise
- $\beta_1, \beta_2 < 0$: Weekend discounts

Three-Regime Specification

The model features three regimes with distinct dynamics:

Regime 0 (Normal - Mean-reverting)

$$\Delta x(t) = -\alpha_0 x(t-1) + \sigma_0 \varepsilon(t), \quad \varepsilon(t) \sim N(0,1) \tag{3}$$

where $\alpha_0 \in (0,1)$ is the mean-reversion rate in normal periods.

Regime +1 (Jump/Spike)

$$\Delta x(t) = \mu_1 + \sigma_1 \varepsilon(t), \quad \varepsilon(t) \sim N(0,1) \tag{4}$$

where μ_1 is the mean jump size.

Regime -1 (Reversion after Spike)

$$\Delta x(t) = -\alpha_{-1} x(t-1) + \sigma_{-1} \varepsilon(t), \quad \varepsilon(t) \sim N(0,1) \tag{5}$$

where $\alpha_{-1} > \alpha_0$ typically, representing faster reversion after spikes.

Markov Transition Structure

The transition matrix (rows = to-state, columns = from-state) is:

$$P = \begin{pmatrix} p(0,0) & 0 & 1 \\ 1 - p(0,0) & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \tag{6}$$

where $p(0,0) = \frac{\exp(p)}{1+\exp(p)}$ is parameterized via logit for estimation.

Questions on Model Fundamentals

Understanding Parameter Roles

Q1 (8%) In the original Huisman and Mahieu paper, what is the key limitation of traditional jump-diffusion models for electricity prices that motivates the regime-switching approach?

Q2 (10%) Explain the economic rationale behind having three separate regimes instead of just two (normal and spike). What specific electricity market characteristic does the reversion regime (-1) capture?

Q3 (8%) The transition matrix in Equation (6) has structural zeros. Explain why:

(a) $P_{-1,0} = 0$ (cannot go from normal directly to reversion)

(b) $P_{+1,+1} = 0$ (spikes cannot last multiple days in this specification)

(c) $P_{0,-1} = 1$ (always return to normal after reversion)

Q4 (9%) The deterministic component only includes weekend effects. Based on the original paper and typical electricity market characteristics, what additional deterministic components might be important for real-world applications? Provide at least two examples with economic justification.

Analysis of Parameter Estimates

Interpreting Parameter Estimates

Q5 (12%) Compare the parameter estimates in Table 1 with the empirical results from Huisman and Mahieu (2003) in their Table 5:

(a) How does the estimated $p(0,0) = 0.8808$ (11.92% spike probability) compare with their findings for APX (0.926), LPX (0.964), and UK (0.950) markets?

(b) The estimated $\alpha_0 = 0.35$ and $\alpha_{-1} = 0.40$. In the original paper, which market showed the largest difference between α_0 and α_{-1}? What might explain this?

(c) Comment on the relative magnitudes of σ_0, σ_1, and σ_{-1} in both cases.

Q6 (10%) Analyze the volatility parameters:

(a) $\sigma_{-1} = 1.1436$ is more than double $\sigma_0 = 0.5283$. What economic or market mechanism might explain higher volatility during reversion periods?

(b) σ_1 (jump volatility) is similar to σ_0 in this simulation. Is this realistic based on the original paper's findings? What would you typically expect?

Q7 (8%) The weekend effects show $\beta_2 < \beta_1$ (Sunday discount larger than Saturday). Is this consistent with the original paper's findings in Table 2? Provide a practical explanation for why Sunday prices might be lower than Saturday prices.

Analyzing Summary Statistics

Q8 (10%) Table 2 shows excess kurtosis of 2.87 and negative skewness of -0.25:

(a) What do these moments tell you about the distribution of electricity price returns?

(b) The original paper (Table 1) shows positive skewness for all markets. Why might our simulation show negative skewness instead? What does this imply about the jump dynamics?

(c) How does the standard deviation of 0.785 compare with the volatility parameters in Table 1?

Analysis of Graphical Outputs

Interpreting Figure 1: Simulation Diagnostics

Q9 (10%) Analyze the price path (top-left panel):

(a) Identify at least three clear spike events in the price series. What is the typical duration from spike onset to return to normal levels?

(b) The legend shows three regimes. Based on the transition structure, what should be the sequence of regimes during a spike event?

(c) Are there periods with clustered spikes? What might cause such clustering in real electricity markets?

Q10 (8%) Analyze the return distribution (top-right panel):

(a) The histogram shows heavy tails. Does this visually confirm the high kurtosis in Table 2?

(b) The mean return line (red dashed) is very close to zero. Is this expected for electricity prices? Why or why not?

Q11 (7%) Analyze the spike frequency by year (bottom-left panel):

(a) Which year had the highest spike frequency? Approximately what percentage of days experienced spikes that year?

(b) Is there any visible trend in spike frequency over the 16-year period?

Interpreting Figure 2: Parameter Diagnostics

Q12 (12%) Figure 2 presents six diagnostic panels. Analyze each panel as follows:

Panel 1: Mean-Reversion Coefficients

(a) The figure shows $\alpha_0 = 0.3500$ (normal) and $\alpha_{-1} = 0.4000$ (reversion). Which is larger? What does this imply about the speed of price adjustment after a spike compared to normal periods?

(b) In the original Huisman and Mahieu (2003) paper, which market (APX, LPX, or UK) showed the largest difference between α_0 and α_{-1}? How does our simulation compare?

Panel 2: Volatility by Regime

(a) The volatilities are $\sigma_0 = 0.5283$ (normal), $\sigma_1 = 0.5311$ (jump), and $\sigma_{-1} = 1.1436$ (reversion). Rank them from lowest to highest.

(b) Why might the reversion regime have the highest volatility? Provide an economic explanation based on market behavior during price corrections.

(c) Is it surprising that σ_1 (jump) is similar to σ_0 (normal)? What would you typically expect based on the literature?

Panel 3: Weekend Effects

(a) The baseline is 3.2000, Saturday is 2.9500, and Sunday is 2.6500. Calculate the absolute and percentage reductions for Saturday and Sunday relative to baseline.

(b) Is the larger Sunday effect ($\beta_2 < \beta_1$) consistent with the original paper's findings in Table 2? Why might Sunday prices be lower than Saturday prices in electricity markets?

Panel 4: Daily Regime Probabilities

(a) The probability of "No Spike" is 88.08%. What is the implied daily spike probability? Convert this to an expected number of spike days per year (assume 252 business days).

(b) Compare this spike frequency with the ranges reported in Huisman and Mahieu (2003): APX (7.4%), LPX (3.6%), UK (5.0%). Is our simulated spike frequency reasonable?

Panel 5: Jump Size Distribution

(a) The panel shows a histogram of jump sizes with a fitted normal density. Based on Table 1 ($\mu_1 = 0.0190$, $\sigma_1 = 0.5311$), what is the approximate range (mean $\pm 2\sigma$) of jump sizes?

(b) Does the histogram appear normally distributed? What features might indicate otherwise, and what alternative distributions could be considered for jumps?

Panel 6: Optimization Diagnostics

(a) The reported values are Log-Likelihood = 48, Iterations = 1, and Convergence = 100.

(b) **Log-Likelihood:** A value of 48 is positive.

- Is a positive log-likelihood value plausible? Under what general condition can the log-likelihood of a continuous model be positive?
- Given the number of observations (n=4195 from Table 2), what would be the average log-likelihood contribution per observation? What does this very small average value suggest about the model fit?

(c) **Iterations:** The optimizer reports 1 iteration.

- Is it credible for a 10-parameter non-linear optimization to converge in a single iteration? What might this imply about the starting parameters or the optimization algorithm's report?
- In a typical scenario using an algorithm like L-BFGS-B, what range of iterations would you expect for this problem?

(d) **Convergence:** The value is 100.

- In common optimization libraries (e.g., 'scipy.optimize.minimize'), what does a convergence code of '0' typically signify? What might a code of '100' or another non-zero number indicate?
- Based on the combination of 'Iterations=1' and 'Convergence=100', what is your overall assessment of the estimation procedure for this run?

(e) What practical steps would you take next to verify the robustness of the parameter estimates in Table 1, given these diagnostics?

End of Exercise. Please hand in your exercise on paper to your instructor.

Table 1: Regime-Switching Model Estimates

Parameter	Symbol	Value
Deterministic Component		
Equilibrium Price Level	μ_0	3.2000
Saturday Effect	β_1	-0.2500
Sunday Effect	β_2	-0.5500
Mean-Reversion (Regime 0 - Normal)		
Mean-Reversion Rate	α_0	0.3500
Normal Volatility	σ_0	0.5283
Jump Regime (Regime +1)		
Jump Mean Size	μ_1	0.0190
Jump Volatility	σ_1	0.5311
Reversion Regime (Regime -1)		
Mean-Reversion Rate	α_{-1}	0.4000
Reversion Volatility	σ_{-1}	1.1436
Markov Transition		
P(Stay in Normal)	$p(0,0)$	0.8808
P(Jump Occurs)	$1-p(0,0)$	0.1192

Table 2: Summary Statistics of Log Returns

Statistic	Value	Unit
Number of Observations	4195	days
Mean Return	-0.000076	log pts
Standard Deviation	0.784759	log pts
Minimum	-4.973307	log pts
Maximum	3.901208	log pts
Skewness	-0.251866	-
Excess Kurtosis	2.868240	-

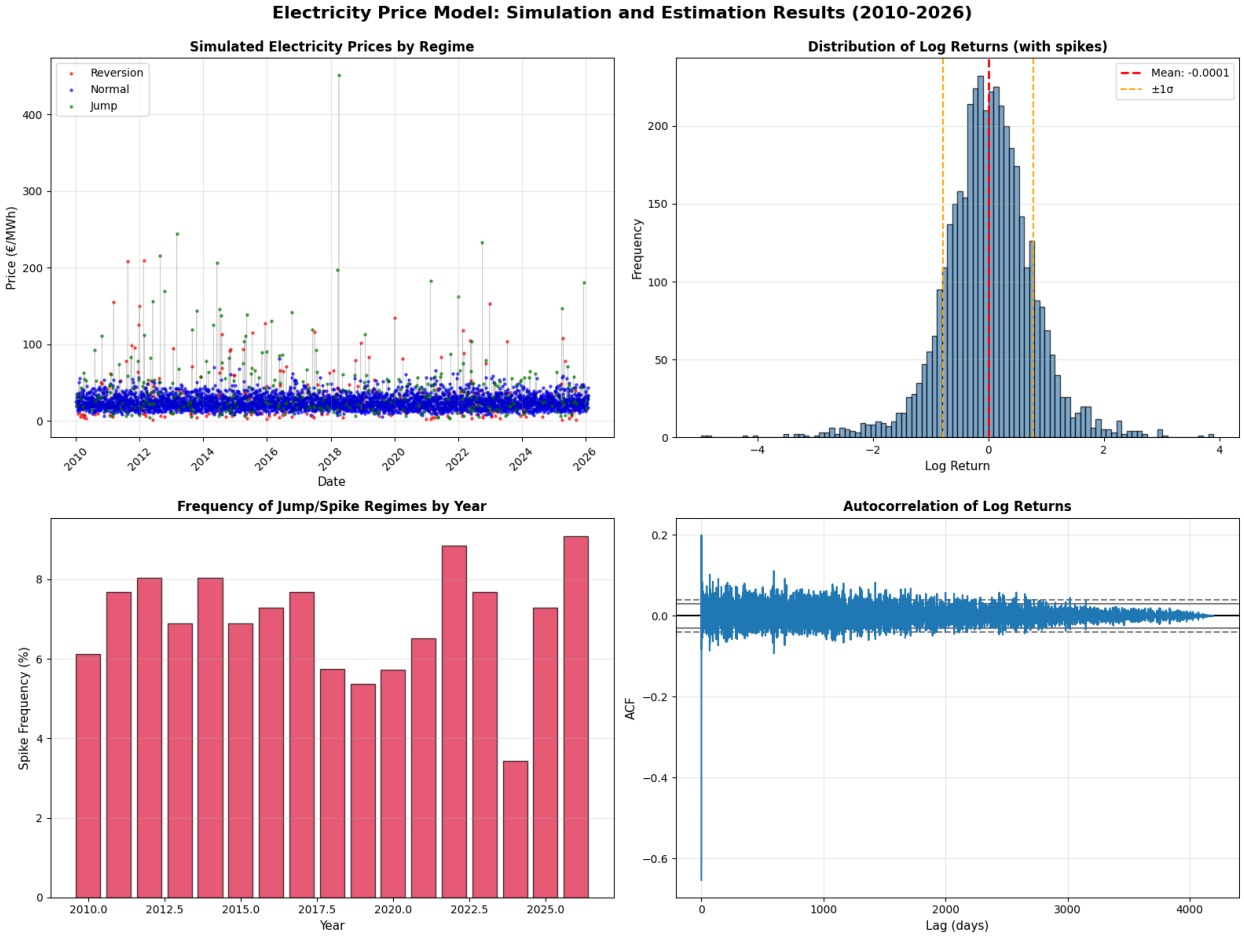

Figure 1: (*from top to bottom and left to right*) Price path by regime; Return distribution; Spike frequency; Autocorrelation

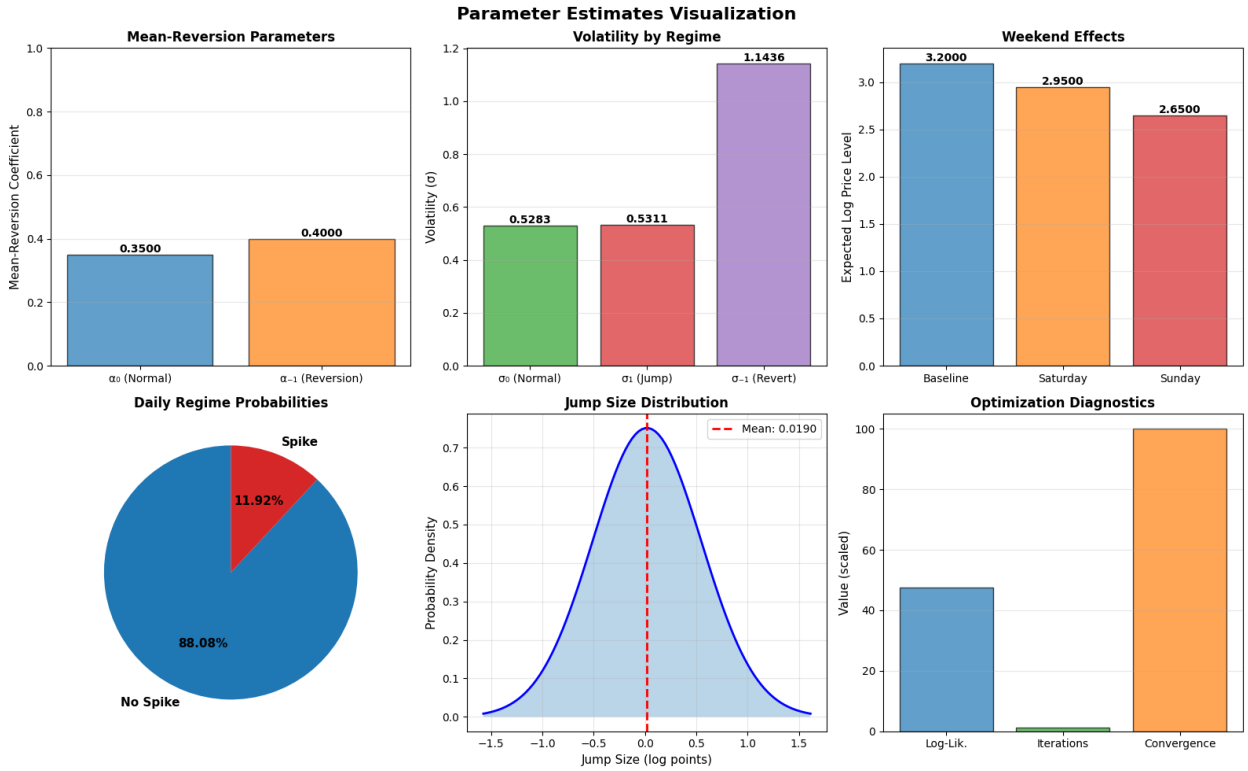

Figure 2: (*from top to bottom and left to right*) Mean-reversion coefficients; Volatility parameters; Weekend effects; Spike statistics; Jump size distribution; Optimization diagnostics